

A Study of the Thermal Behaviour of Different Ground Media in Response to the Polar Cold Front in Antarctica

A. CHATURVEDI and R. ASTHANA

Geological Survey of India, Antarctica Division
Faridabad - 121001

Abstract

Thermal probing was attempted in different ground media, with the help of thermistors. For three media (ice, firn and permafrost) day-to-day variations have been plotted for about ten months. Comparative inferences about the conductivity, sensitivity and thermal-amplitude of these media have been drawn. Also, monthly thermal profiles for eight months have been plotted for a closed crevasse, showing the variations inside a crevasse through the polar winter.

Introduction

The cold regions at the poles act as heat sinks for the atmosphere of the earth and maintain an equilibrium. Antarctica, the highest continent of the world, receives 40% less radiation than the equator. Since, 98% of the continent is covered by ice and snow, most of this radiation is reflected back due to high albedo. In general, the mean annual temperature varies from -30°C to -70°C on the polar ice sheet; while around coastal areas, it ranges from -2°C to -40°C .

The arrival of the polar cold front during polar winter and the warm front during polar summer, induce significant thermal changes in all the media. To find out the exact nature of these changes and to compare the thermal behaviour of different ground media, this study was undertaken.

Maitri, the Indian Antarctic station ($70^{\circ}45'39''\text{S}$ latitude and $11^{\circ}44'48''\text{E}$ longitude), is situated in Schirmacher Oasis of Central Dronning Maud Land in East Antarctica. The southern margin of the oasis is flanked all along by a glacial front of polar continental "blue" ice. The glacial front stands in the form of a 10 to 30 m high escarpment; but at places, sloping ramps are made by snow, deposited by blizzards. This is a recurring phenomenon and as the older snow gets metamorphosed, it gives rise to firn. In the first study, about half-a-km

south of Maitri, three media were selected for comparison – blue ice, firn and permafrost. The site of the study is at an altitude of about 150 m above the MSL.

In the second part of the study, a crevasse, 30 m deep, was located near nunatak Veteheia, at 70°47'05"S and 11°37'10"E. It was selected for its covered nature and relatively easy accessibility. The objective of the study was to find out the thermal changes that occur inside a crevasse, in response to the polar winter. The site of these observations is at an altitude of about 300 m above the MSL.

Equipment

For the first study, four thermistors (resistance thermal detectors) were used, each with a range of –40°C to +20°C and with two metres long teflon-coated wire attached to it. For the second study, a single such thermistor was utilised, but it had a 50 m long teflon-coated wire attached to it. This wire was coiled on a small pulley-system, for lowering or hoisting the thermistor. With the help of a digital multimeter, all these thermistors were calibrated at different negative temperatures. After calibration, temperatures were deduced from measured resistances by the following relationship :

$$X_2 - X = \frac{(X_2 - X_1)(Y_2 - Y)}{(Y_2 - Y_1)}$$

where, X_1 and X_2 are calibrated temperatures in °C for Y_1 and Y_2 resistances in ohms; and X is the temperature to be found in °C, when Y is the measured resistance in ohms. The accuracy of the thermistors was + 0.1°C.

Part A : Comparative Thermal Study of Ice-Firn-Permafrost

In each of the three media, one shallow borehole was made upto a depth of one metre. The calibrated thermistors were installed in horizontal positions, at the bottom of the boreholes and these holes were carefully filled with the original scooped-out medium, leaving the teflon-wires peeping outside for measurements. The fourth thermistor was used for measuring ice-surface temperature on the glacier.

At the time of these observations, corresponding air-temperatures were also noted. These were provided by the sensor of India Meteorological Department, installed at a height of 1.5 m above the ground, north of Maitri station.

Observations

For about ten months, from May 1993 to February 1994, whenever the weather permitted, observations were taken on a day-to-day basis. At times, more than one reading were taken in a day, to check the effect of diurnal fluctuations. A total of 177 observations have been recorded. These are presented in Table 1.

The glacier became unapproachable due to partial surface-melting after 25th December, 1993. Such conditions persisted till the second half of January. It became accessible on 22 January, 1994. So, there are no observations between 25.12.93 - 22.1.94.

Discussion

When the observations started in the month of May 1993, permafrost was the coldest medium starting at a temperature of -13°C , followed by blue ice at -12°C and firn was around -10°C . Later, during this month, the ice-surface temperature touched a low of -26°C ; which was reflected in blue ice dipping to -16°C . Firn also responded to reach -12°C , but permafrost was almost unaffected by it (Fig. 1).

In June, the lowest ice-surface temperature was -29°C . Blue ice and firn remained similar to May, around -16°C and -13°C respectively. Permafrost

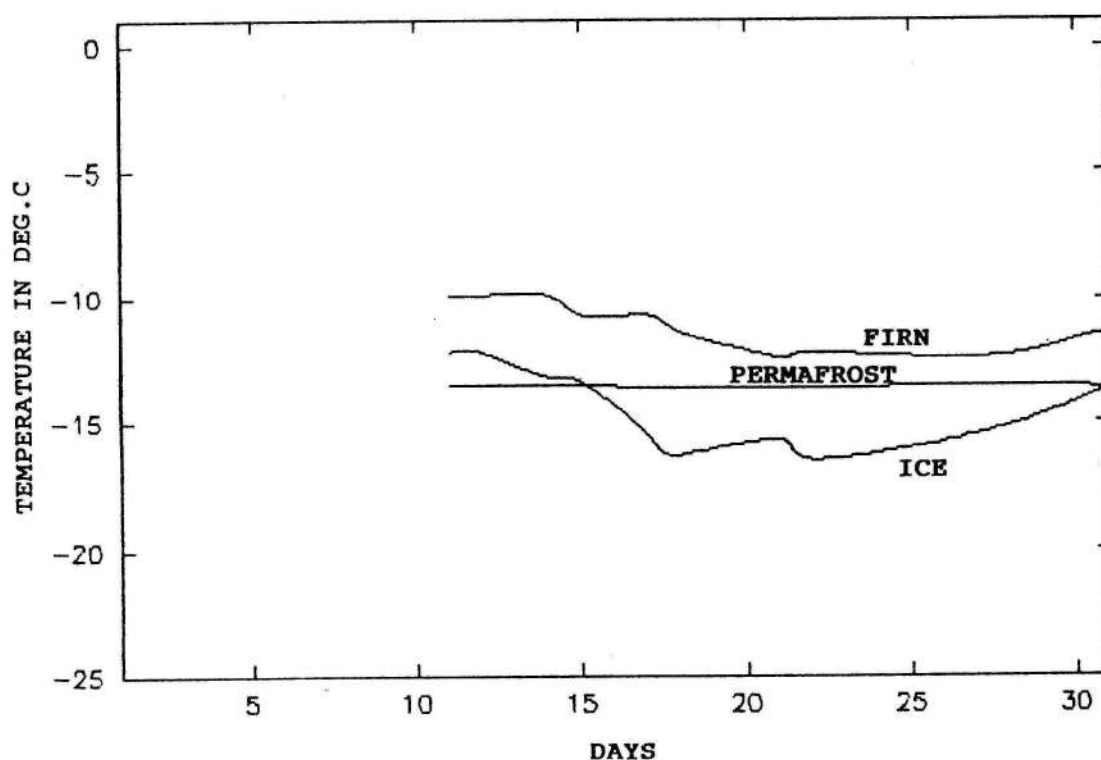


Fig. 1. Thermal plots for ice-firn-permafrost for May 1993

Table 1: Day-to-day thermal observations from May 1993 to February 1994

Date	Time (in GMT)	Wind Speed (km/h)	Blue Ice-T (°C)	Firn Temp (°C)	Perma- forst-T (°C)	Surface Ice-T (°C)	Air Temp (°C)
11-May-93	1200	35	-12.2	-9.9	-13.5	-15.1	-12.5
12-May-93	1200	30	-12.2	-9.9	-13.5	-16.1	-14.5
14-May-93	0900	35	-13.2	-9.9	-13.5	-16.8	-16.0
15-May-93	0900	0	-13.4	-10.7	-13.5	-26.4	-23.5
15-May-93	1500	10	-13.4	-10.7	-13.5	-24.9	-22.1
17-May-93	0900	55	-15.5	-10.7	-13.6	-19.6	-18.0
18-May-93	0900	55	-16.3	-11.4	-13.6	-18.3	-17.0
21-May-93	0900	0	-15.7	-12.4	-13.6	-24.1	-23.0
21-May-93	2100	15	-15.8	-12.4	-13.6	-22.5	-20.0
22-May-93	1200	0	-16.5	-12.2	-13.6	-21.4	-19.0
27-May-93	1200	35	-15.5	-12.4	-13.5	-10.3	-9.0
30-May-93	0900	30	-14.2	-11.7	-13.5	-14.7	-11.0
31-May-93	0900	20	-13.9	-11.5	-13.7	-9.2	-7.0
31-May-93	1500	0	-13.7	-11.4	-13.7	-10.8	-8.0
01-Jun-93	1200	30	-13.4	-11.2	-13.8	-14.7	-12.1
02-Jun-93	0900	30	-13.4	-11.2	-13.9	-12.9	-10.0
03-Jun-93	0900	0	-13.9	-11.4	-13.9	-17.8	-15.5
04-Jun-93	0900	45	-13.5	-11.4	-14.0	-13.6	-11.5
05-Jun-93	0900	35	-13.9	-10.9	-13.8	-20.7	-16.1
06-Jun-93	0900	0	-14.5	-11.4	-13.7	-18.9	-17.0
07-Jun-93	0900	35	-15.2	-11.7	-13.5	-23.0	-19.5
08-Jun-93	0900	30	-15.7	-11.9	-13.7	-26.7	-22.0
15-Jun-93	0900	65	-13.5	-11.4	-13.6	-16.5	-13.4
16-Jun-93	0900	0	-13.2	-11.2	-13.9	-22.5	-19.0
17-Jun-93	1200	20	-14.5	-11.7	-13.9	-24.6	-24.0
18-Jun-93	1200	65	-14.7	-12.0	-14.0	-17.8	-14.6
19-Jun-93	1200	35	-15.5	-12.5	-13.9	-17.3	-15.0
22-Jun-93	0900	0	-15.7	-12.7	-13.7	-26.5	-23.4
22-Jun-93	1200	0	-15.7	-12.7	-13.7	-26.6	-22.0
23-Jun-93	0900	30	-16.2	-12.7	-13.6	-23.8	-20.5
23-Jun-93	1200	35	-16.2	-12.7	-13.6	-21.9	-18.5

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Table 1: Contd.

Date	Time (in GMT)	Wind Speed (km/h)	Blue Ice-T (°C)	Firn Temp (°C)	Perma- forst-T (°C)	Surface Ice-T (°C)	Air Temp (°C)
27-Jun-93	0900	20	-15.7	-12.7	-13.7	-19.9	-17.0
28-Jun-93	1200	20	-15.8	-12.9	-13.6	-23.0	-18.0
29-Jun-93	0900	35	-16.2	-13.0	-13.7	-24.1	-22.0
30-Jun-93	1200	0	-16.7	-13.2	-13.6	-29.8	-27.0
01-Jul-93	1200	20	-17.7	-13.5	-13.7	-28.2	-27.5
03-Jul-93	1200	55	-17.7	-14.2	-13.4	-17.6	-12.0
07-Jul-93	1200	75	-16.0	-13.7	-13.6	-18.9	-14.5
08-Jul-93	1200	20	-15.2	-13.5	-13.4	-23.0	-21.0
10-Jul-93	1200	55	-16.2	-13.2	-13.6	-11.8	-8.0
11-Jul-93	1200	45	-15.8	-13.2	-13.6	-21.7	-15.6
12-Jul-93	1200	65	-16.5	-13.5	-13.8	-30.3	-26.5
13-Jul-93	1200	65	-17.5	-13.5	-13.5	-32.4	-26.6
14-Jul-93	1200	20	-18.2	-13.6	-13.4	-28.5	-27.0
15-Jul-93	0900	0	-18.8	-13.6	-13.4	-35.8	-29.0
15-Jul-93	1500	15	-18.8	-13.6	-13.4	-31.3	-26.0
16-Jul-93	0900	45	-19.2	-13.6	-13.4	-29.8	-23.9
17-Jul-93	0900	35	-19.8	-13.7	-13.2	-24.3	-22.1
19-Jul-93	0900	35	-19.9	-13.9	-13.2	-25.6	-21.0
20-Jul-93	0900	35	-20.0	-14.0	-13.1	-32.4	-30.0
21-Jul-93	0900	55	-20.2	-14.2	-13.3	-26.8	-24.0
23-Jul-93	1200	30	-19.9	-14.2	-13.1	-21.5	-17.2
24-Jul-93	0900	30	-19.2	-13.7	-13.1	-27.4	-24.2
25-Jul-93	0900	0	-19.5	-13.9	-13.1	-25.1	-24.2
25-Jul-93	1800	15	-19.5	-13.9	-13.1	-24.6	-22.0
27-Jul-93	0900	20	-20.0	-14.1	-13.4	-20.4	-17.0
28-Jul-93	0900	30	-20.0	-14.1	-13.4	-26.4	-24.2
29-Jul-93	1200	20	-20.1	-14.2	-13.4	-31.3	-27.6
30-Jul-93	0900	30	-20.5	-14.3	-13.4	-28.7	-23.2
31-Jul-93	1200	75	-21.5	-14.3	-13.4	-23.0	-20.0
04-Aug-93	0900	75	-18.7	-14.0	-13.3	-8.9	-7.4
05-Aug-93	1200	65	-18.0	-13.9	-13.1	-14.2	-11.8

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Table 1: *Contd.*

Date	Time (in GMT)	Wind Speed (km/h)	Blue Ice-T (°C)	Firn Temp (°C)	Perma- forst-T (°C)	Surface Ice-T (°C)	Air Temp (°C)
06-Aug-93	0900	35	-17.5	-13.9	-13.1	-13.7	-12.0
07-Aug-93	1200	85	-17.0	-13.8	-13.1	-12.9	-11.6
08-Aug-93	1200	75	-16.7	-13.6	-13.1	-13.1	-10.2
09-Aug-93	1200	55	-16.6	-13.5	-13.1	-19.4	-15.6
10-Aug-93	0900	35	-16.7	-13.5	-12.9	-20.1	-18.0
11-Aug-93	0900	35	-17.2	-13.6	-12.9	-23.8	-21.4
13-Aug-93	1200	65	-17.7	-13.5	-13.0	-17.6	-14.0
15-Aug-93	1200	55	-17.1	-13.3	-13.0	-11.3	-7.9
16-Aug-93	0900	45	-16.6	-13.2	-13.0	-15.3	-13.0
17-Aug-93	1200	45	-16.5	-13.2	-13.0	-17.2	-14.6
18-Aug-93	0600	35	-16.5	-13.2	-13.0	-19.5	-17.5
22-Aug-93	1200	75	-17.9	-13.5	-12.9	-12.1	-10.4
23-Aug-93	1200	20	-17.7	-13.4	-12.9	-21.8	-18.5
24-Aug-93	1500	30	-17.5	-13.4	-12.9	-26.9	-23.3
26-Aug-93	0600	20	-18.9	-13.7	-13.1	-21.7	-18.8
26-Aug-93	1500	0	-19.0	-13.7	-13.1	-25.8	-22.0
28-Aug-93	0600	0	-19.4	-13.9	-13.2	-26.9	-24.6
28-Aug-93	1500	0	-19.4	-13.9	-13.1	-28.5	-27.6
28-Aug-93	2100	0	-19.6	-13.9	-13.1	-32.2	-28.2
29-Aug-93	0600	0	-20.0	-14.0	-13.1	-34.6	-30.0
30-Aug-93	1200	0	-20.3	-14.0	-13.1	-23.0	-22.4
31-Aug-93	0600	20	-20.1	-14.0	-13.1	-27.1	-24.0
31-Aug-93	1500	0	-20.1	-14.0	-13.1	-29.4	-26.0
02-Sep-93	1500	75	-20.3	-14.1	-13.2	-16.4	-12.3
03-Sep-93	1500	75	-19.2	-14.4	-13.1	-16.8	-13.8
04-Sep-93	1500	75	-18.8	-14.5	-13.1	-14.1	-11.5
05-Sep-93	0900	0	-18.7	-14.5	-13.3	-15.5	-13.3
05-Sep-93	1800	5	-18.7	-14.5	-13.3	-18.7	-16.0
07-Sep-93	1200	75	-18.8	-14.5	-13.4	-12.4	-12.0
08-Sep-93	0900	35	-17.9	-14.5	-13.4	-16.3	-15.8
09-Sep-93	0900	35	-17.5	-14.4	-13.4	-25.2	-22.0

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Table 1: *Contd.*

Date	Time (in GMT)	Wind Speed (km/h)	Blue Ice-T (°C)	Firn Temp (°C)	Perma- forst-T (°C)	Surface Ice-T (°C)	Air Temp (°C)
11-Sep-93	1200	20	-17.7	-14.3	-13.4	-22.9	-19.5
12-Sep-93	1500	35	-18.3	-14.4	-13.3	-12.8	-11.5
14-Sep-93	0900	0	-17.7	-14.3	-13.4	-18.3	-18.0
14-Sep-93	1500	0	-17.7	-14.3	-13.4	-21.2	-19.8
14-Sep-93	2100	10	-17.8	-14.3	-13.4	-20.8	-18.6
19-Sep-93	1200	45	-16.6	-14.2	-13.4	-14.4	-13.0
20-Sep-93	1500	0	-16.5	-14.1	-13.4	-17.3	-13.5
21-Sep-93	1500	20	-16.5	-14.0	-13.4	-16.7	-13.2
24-Sep-93	1200	30	-16.2	-14.0	-13.4	-14.1	-12.8
25-Sep-93	0900	20	-15.7	-14.0	-13.5	-15.7	-14.8
27-Sep-93	1500	55	-15.5	-14.0	-13.4	-15.4	-13.0
28-Sep-93	1800	55	-15.6	-13.9	-13.4	-14.4	-10.4
29-Sep-93	1200	20	-15.5	-13.8	-13.4	-13.1	-11.0
30-Sep-93	0900	55	-15.3	-13.6	-13.4	-12.1	-10.1
01-Oct-93	1200	30	-15.0	-13.6	-13.4	-11.6	-9.5
02-Oct-93	0600	0	-14.7	-13.5	-13.4	-15.6	-13.2
02-Oct-93	1500	0	-14.6	-13.5	-13.4	-12.8	-11.0
02-Oct-93	2100	0	-14.6	-13.5	-13.4	-16.3	-13.0
03-Oct-93	1500	20	-14.2	-13.6	-13.4	-12.6	-11.3
06-Oct-93	1800	55	-13.7	-13.6	-13.4	-13.3	-10.6
07-Oct-93	1500	20	-13.3	-13.6	-13.4	-13.4	-11.0
08-Oct-93	0600	0	-13.1	-13.6	-13.4	-21.4	-19.4
09-Oct-93	1800	45	-12.9	-13.6	-13.4	-16.1	-14.0
10-Oct-93	0900	0	-12.4	-13.5	-13.4	-13.7	-11.0
10-Oct-93	1800	0	-12.4	-13.5	-13.4	-18.8	-15.0
13-Oct-93	0600	55	-12.4	-13.2	-13.4	-17.8	-16.0
16-Oct-93	1200	55	-11.9	-13.0	-13.3	-6.3	-4.0
19-Oct-93	0600	55	-11.4	-12.8	-13.3	-13.1	-11.0
21-Oct-93	0600	0	-11.3	-12.7	-13.3	-17.3	-14.2
21-Oct-93	1800	0	-11.3	-12.7	-13.3	-15.9	-12.0
22-Oct-93	0600	55	-11.4	-12.7	-13.1	-14.2	-12.0

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Table 1: *Contd.*

Date	Time (in GMT)	Wind Speed (km/h)	Blue Ice-T (°C)	Firn Temp (°C)	Perma- forst-T (°C)	Surface Ice-T (°C)	Air Temp (°C)
24-Oct-93	0900	30	-11.4	-12.7	-13.1	-9.7	-8.8
26-Oct-93	1500	30	-10.7	-12.7	-13.1	-9.7	-9.5
27-Oct-93	1800	45	-10.8	-12.7	-13.1	-13.1	-10.6
29-Oct-93	0900	45	-10.3	-12.5	-13.1	-7.5	-6.4
30-Oct-93	0600	0	-10.0	-12.4	-13.0	-11.9	-9.8
30-Oct-93	1500	0	-9.9	-12.3	-13.0	-10.7	-8.8
01-Nov-93	1800	85	-9.7	-12.3	-13.0	-10.7	-9.0
04-Nov-93	1500	20	-9.3	-12.1	-12.9	-9.4	-6.2
05-Nov-93	1200	55	-9.3	-12.1	-12.9	-10.2	-8.0
08-Nov-93	1500	30	-8.8	-11.9	-12.9	-6.2	-2.0
09-Nov-93	0900	20	-8.7	-11.9	-12.9	-6.4	-2.5
10-Nov-93	1200	20	-7.6	-11.7	-12.8	-4.7	-3.5
11-Nov-93	0900	10	-7.2	-11.6	-12.8	-7.6	-7.0
12-Nov-93	1800	65	-6.8	-11.5	-12.7	-5.2	0.0
13-Nov-93	2100	75	-5.6	-11.4	-12.7	-7.3	-3.2
15-Nov-93	0600	55	-4.7	-11.4	-12.7	-6.5	-3.0
16-Nov-93	0300	10	-4.7	-11.4	-12.7	-9.8	-7.0
16-Nov-93	1200	10	-4.6	-11.4	-12.6	-7.6	-5.5
16-Nov-93	2100	0	-4.6	-11.4	-12.6	-11.8	-9.5
17-Nov-93	2100	35	-5.1	-11.1	-12.6	-12.0	-8.5
19-Nov-93	2100	45	-5.1	-11.1	-12.6	-11.0	-8.8
21-Nov-93	2100	20	-5.1	-11.1	-12.6	-10.7	-5.9
23-Nov-93	0600	0	-4.7	-11.1	-12.6	-6.8	-4.5
23-Nov-93	2100	0	-4.6	-11.1	-12.6	-5.4	-2.8
24-Nov-93	2100	55	-4.2	-11.0	-12.5	-4.7	-1.8
26-Nov-93	0600	5	-3.5	-10.8	-12.5	-5.6	-2.0
26-Nov-93	2100	0	-3.4	-10.8	-12.5	-7.3	-3.0
28-Nov-93	2100	35	-3.1	-10.3	-12.5	-8.9	-5.0
29-Nov-93	2100	35	-2.9	-10.3	-12.4	-8.6	-4.0
01-Dec-93	2400	35	-2.8	-10.3	-12.3	-7.5	-5.0
02-Dec-93	2400	30	-2.8	-10.0	-12.3	-8.3	-5.4

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Table 1: *Contd.*

Date	Time (in GMT)	Wind Speed (km/h)	Blue Ice-T (°C)	Firn Temp (°C)	Perma- frost-T (°C)	Surface Ice-T (°C)	Air Temp (°C)
04-Dec-93	2100	55	-2.8	-10.1	-12.3	-8.1	-4.0
07-Dec-93	2100	55	-2.8	-10.0	-12.3	-3.8	1.0
11-Dec-93	1500	35	-2.1	-9.4	-12.1	-2.1	0.4
14-Dec-93	1500	65	-2.1	-9.3	-11.8	-2.6	0.2
15-Dec-93	1800	65	-2.0	-9.1	-11.8	-1.6	0.2
17-Dec-93	1500	20	-1.8	-8.9	-11.7	-1.0	0.0
18-Dec-93	0300	0	-1.7	-8.8	-11.6	-3.8	-2.0
18-Dec-93	0900	0	-1.6	-8.8	-11.6	-2.7	-1.1
18-Dec-93	1200	0	-1.6	-8.8	-11.6	-0.8	-0.1
18-Dec-93	2100	10	-1.6	-8.8	-11.6	-2.8	-1.2
19-Dec-93	2100	35	-1.6	-7.4	-11.5	-4.6	-1.5
22-Dec-93	2400	30	-1.3	-0.1	-2.2	-3.6	-0.5
23-Dec-93	2100	55	-1.2	-0.1	-2.2	-3.8	1.1
24-Dec-93	2100	30	-1.1	-0.1	-2.2	-1.5	2.0
25-Dec-93	2100	20	-1.1	-0.1	-2.2	-3.1	1.0
22-Jan-94	2400	0	-0.3	-0.6	-3.2	-4.2	-1.4
24-Jan-94	1800	0	-0.3	-0.6	-3.2	-1.1	2.0
30-Jan-94	2100	0	-0.3	-0.6	-3.2	-6.8	-3.9
02-Feb-94	2400	55	-0.5	-0.6	-3.2	-9.8	-6.0
09-Feb-94	2400	0	-1.6	-0.6	-3.2	-9.4	-5.0
11-Feb-94	2100	0	-1.6	-0.6	-3.2	-8.3	-4.8
14-Feb-94	1200	30	-1.6	-0.6	-3.2	-4.6	-3.0

touched -14°C and this was the lowest temperature recorded in this medium, during the entire year (Fig. 2).

July was the coldest month, with the ice-surface temperature recording a lowest of -35°C . Blue ice was quick to respond and remained around -20°C ; touching a low of -22°C , the lowest recorded in this medium during the year. Firn dipped to -14°C , while permafrost remained stable below -13°C (Fig. 3).

August was only a bit less cold than July and blue ice remained around 17°C , touching a low of -20°C . Firn and permafrost were stable around -14°C and -13°C , respectively (Fig. 4).

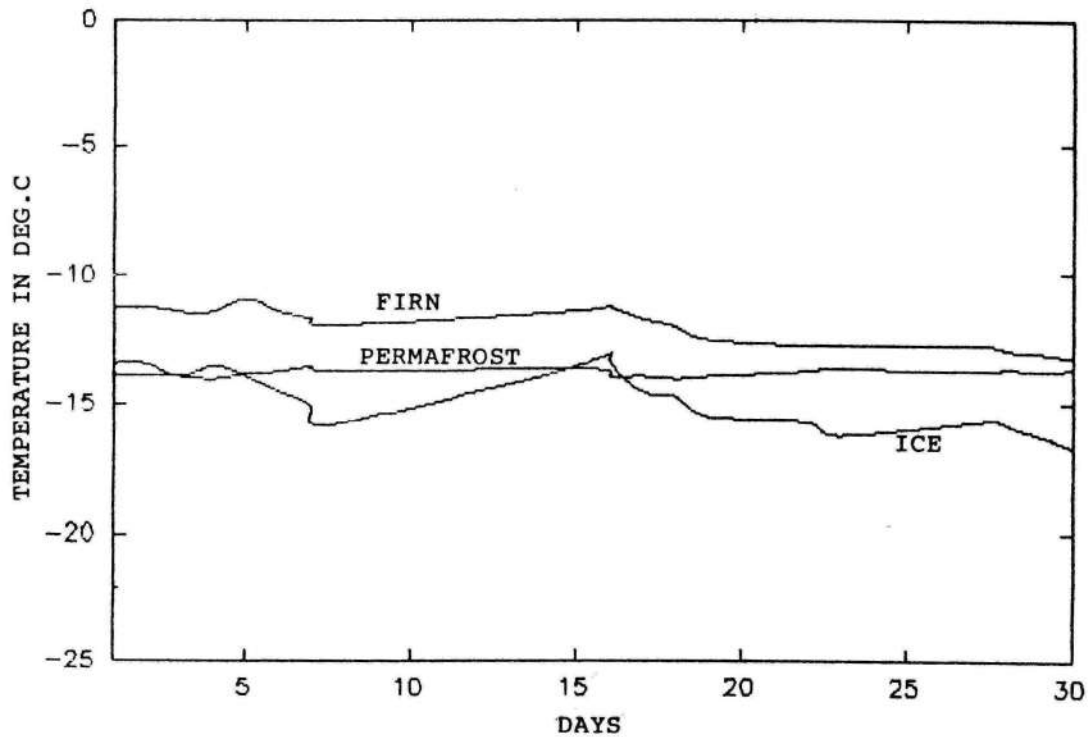


Fig. 2. Thermal plots for ice-firn-permafrost for June 1993

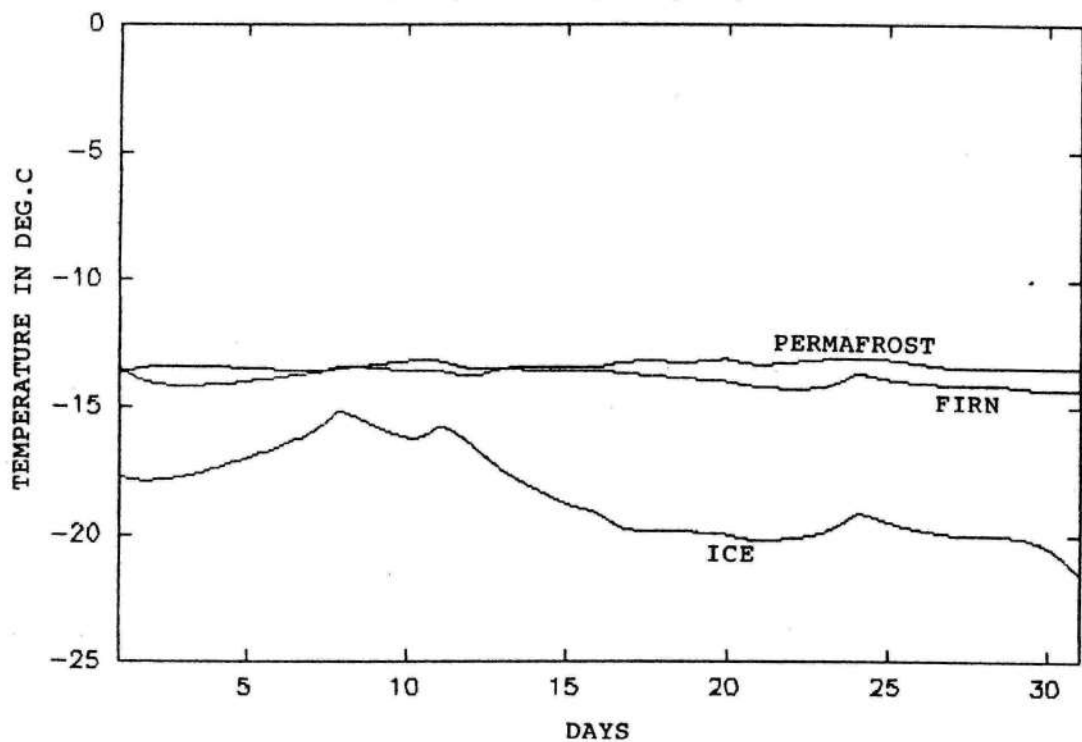


Fig. 3. Thermal plots for ice-firn-permafrost for July 1993

Surface temperatures started rising in September and blue ice responded by rising from -20°C to -15°C . After an initial warming, firn and permfrost both recorded consistent dipping trends; with firn touching -14.5°C , the lowest temperature recorded in the year in this medium (Fig. 5).

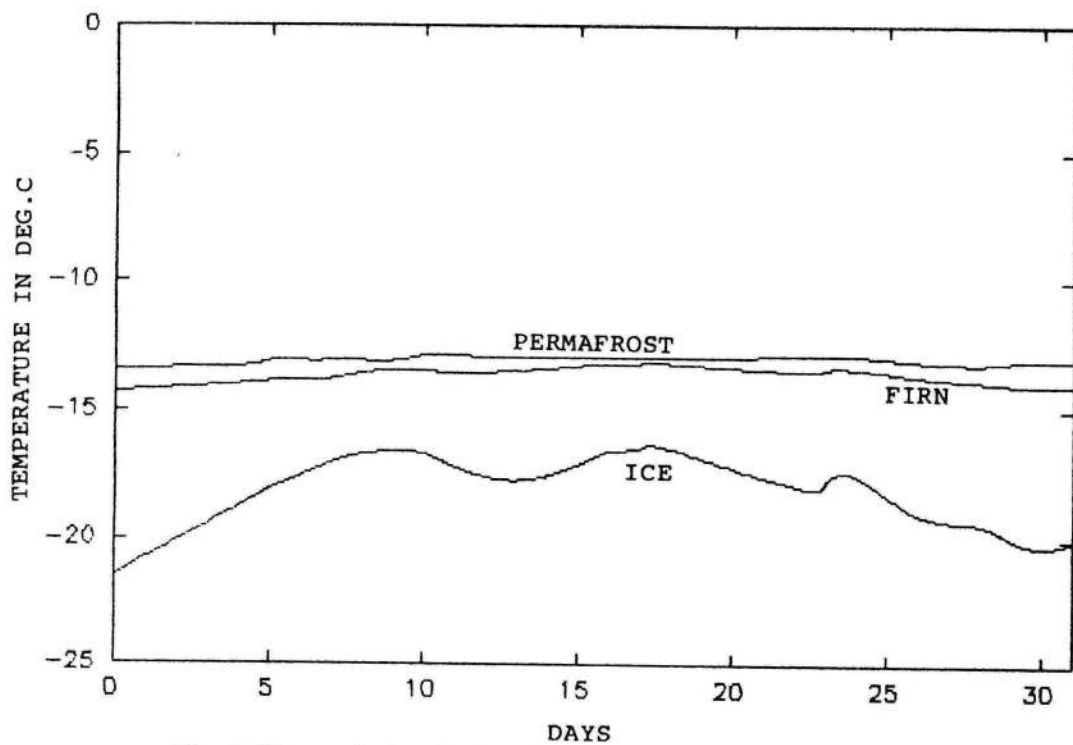


Fig. 4. Thermal plots for ice-firn-permafrost for August 1993

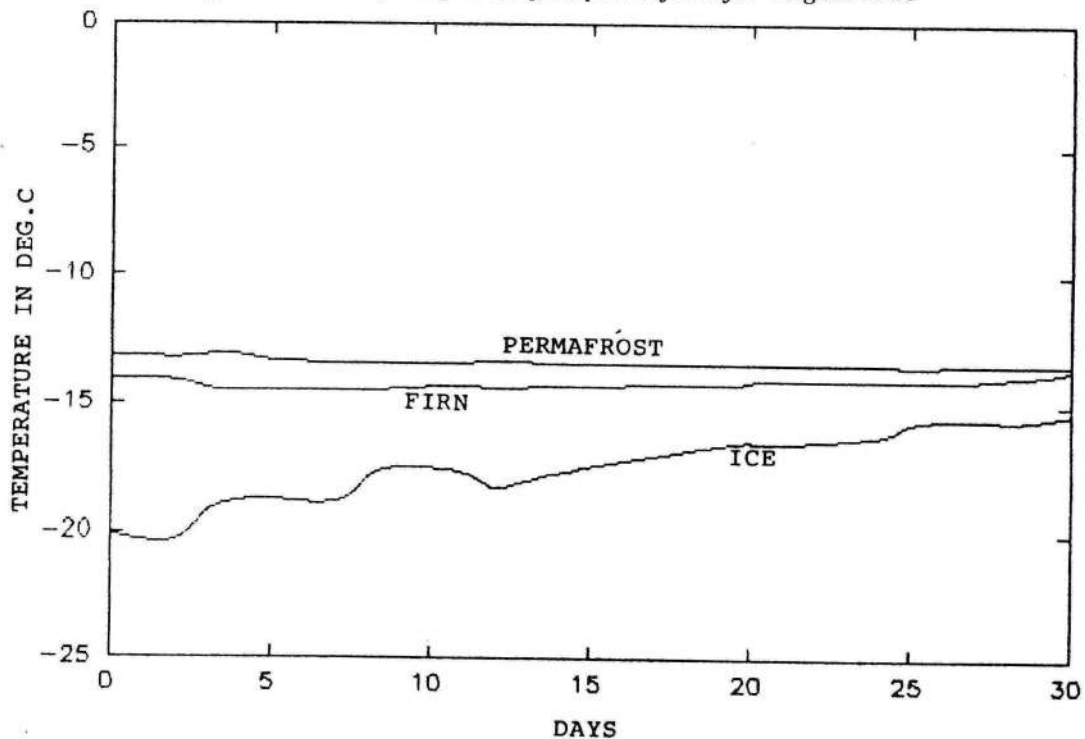


Fig. 5. Thermal plots for ice-firn-permafrost for September, 1993

This divergent phenomenon, (the consistent dipping in firn and permafrost, while surface temperatures and blue ice were showing rising trends), may be explained as a typical delayed response in firn and permafrost. The last week of August was uninterrupted cold, due to absence of strong winds, and the

ice-surface temperature ranged from -22°C to -34°C . Its response was registered immediately in blue ice, but in firn and permafrost, it was recorded in the first week of September.

In October, the average surface temperature was around -13°C and blue ice steadily rose from -15°C to -10°C . Firn and permafrost also registered a small rise, touching -12°C and -13°C , respectively (Fig. 6).

The polar summer was firmly set by November and the average surface temperature was around -6°C . Blue ice continued the steady rise from -10°C to -3°C . Firn and permafrost rose to reach -10°C and -12°C , respectively (Fig. 7).

In December, the first positive air-temperatures were recorded, but the ice-surface touched upto -1°C . Blue ice steadily climbed from -3°C to -1°C . Firn (-7°C) and permafrost (-11°C) rose slowly till 19 December. On December 22, a sudden jump was recorded in both these media, taking them to 0°C and -2°C , respectively. This may be explained by surface-melting and the resultant percolation of melt-water below (Fig. 8).

After 25 December, the surface of the glacier started melting and the observation sites became inaccessible.

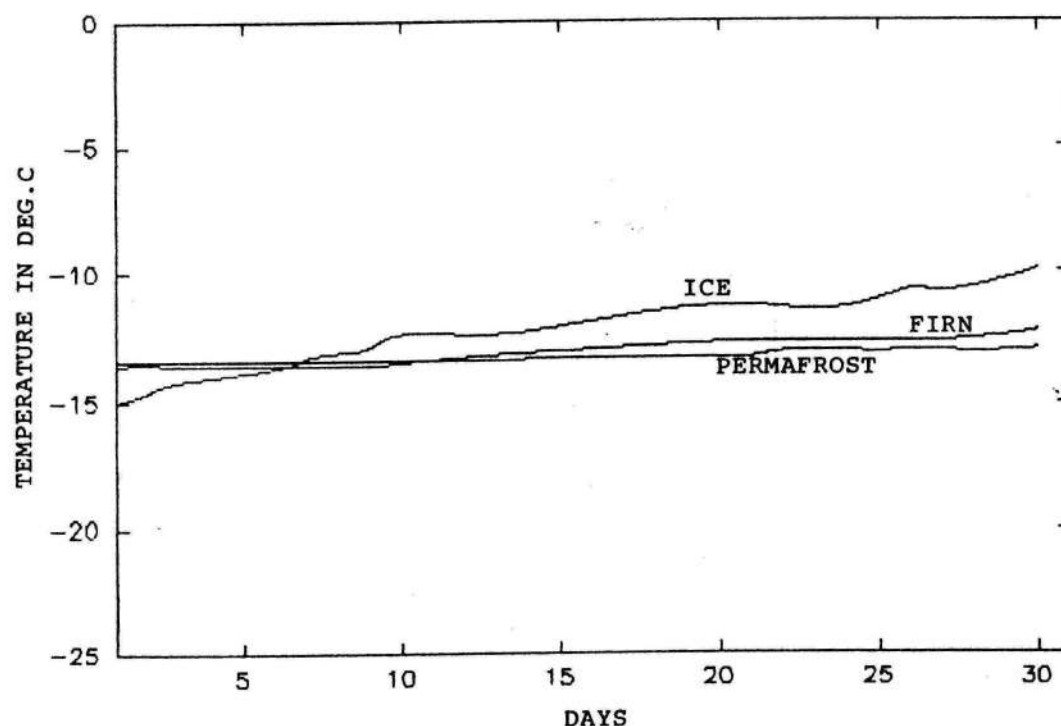


Fig. 6. Thermal plots for ice-firn-permafrost for October 1993

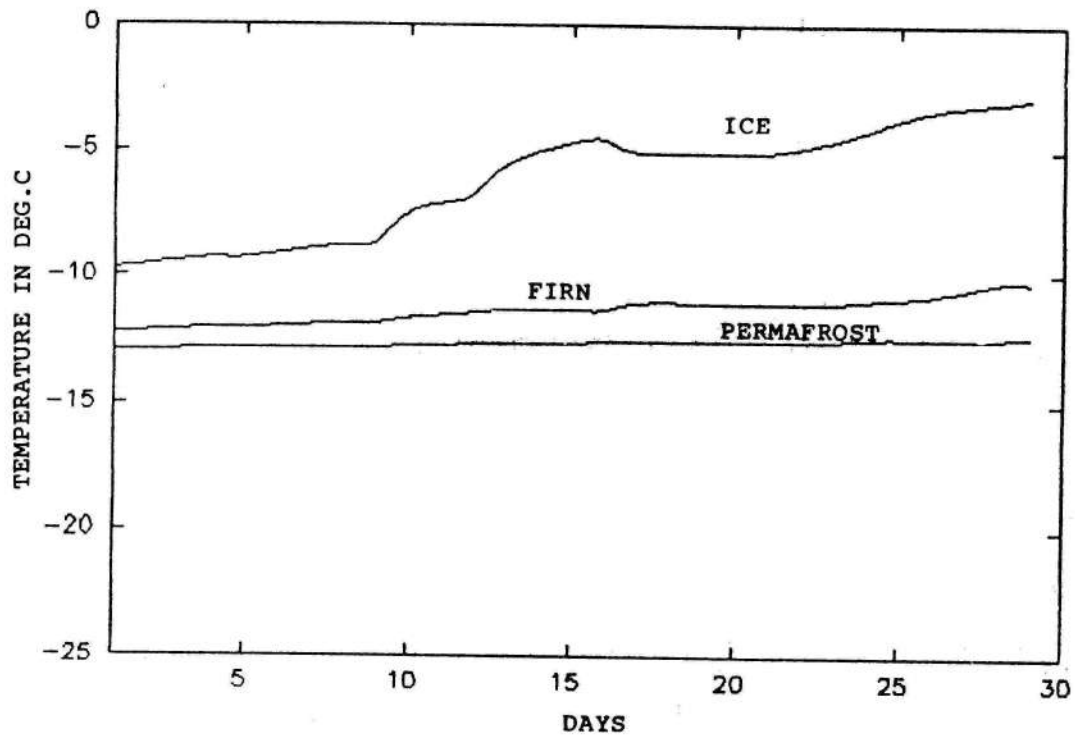


Fig. 7. Thermal plots for ice-firn-permafrost for November 1993

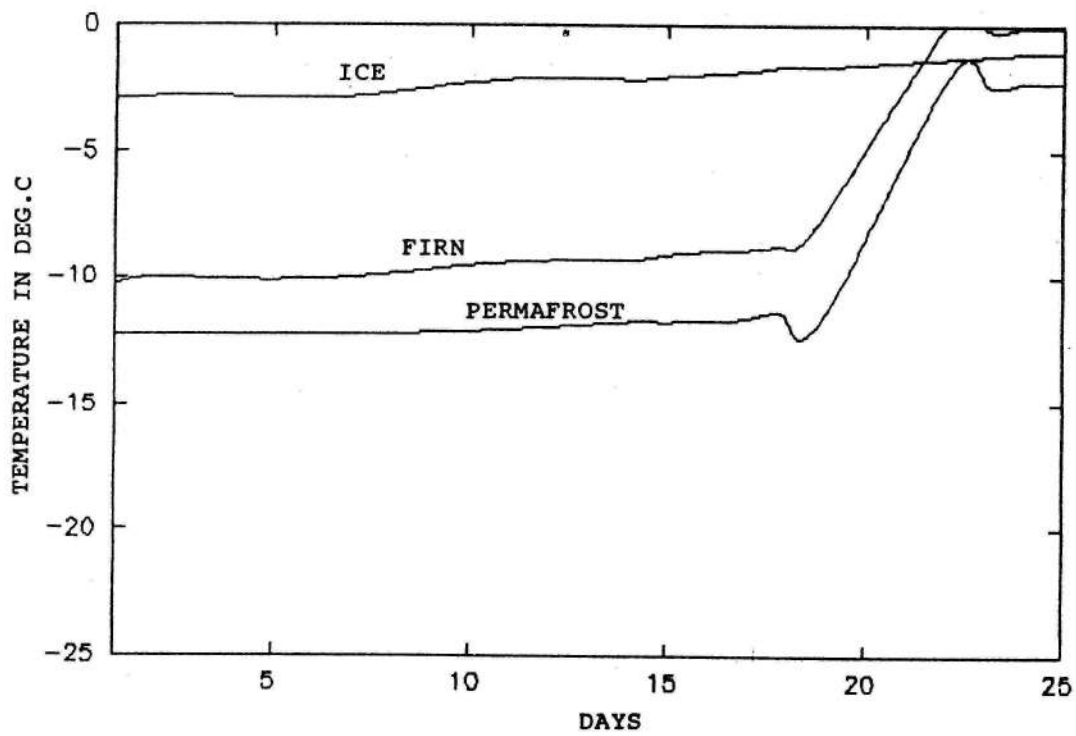


Fig. 8. Thermal plots for ice-firn-permafrost for December 1993

The next observation was possible on 22 January, 1994. By now, all the media were stable; blue ice and firn around 0°C and permafrost around -3°C . (Fig. 9).

In February, with the polar summer reaching its end, ice surface temperatures again started dipping to as low as -9°C . The first to respond was again blue ice, dipping to -1.5°C ; while firn and permafrost remained stable at 0°C and -3°C , respectively (Fig. 10).

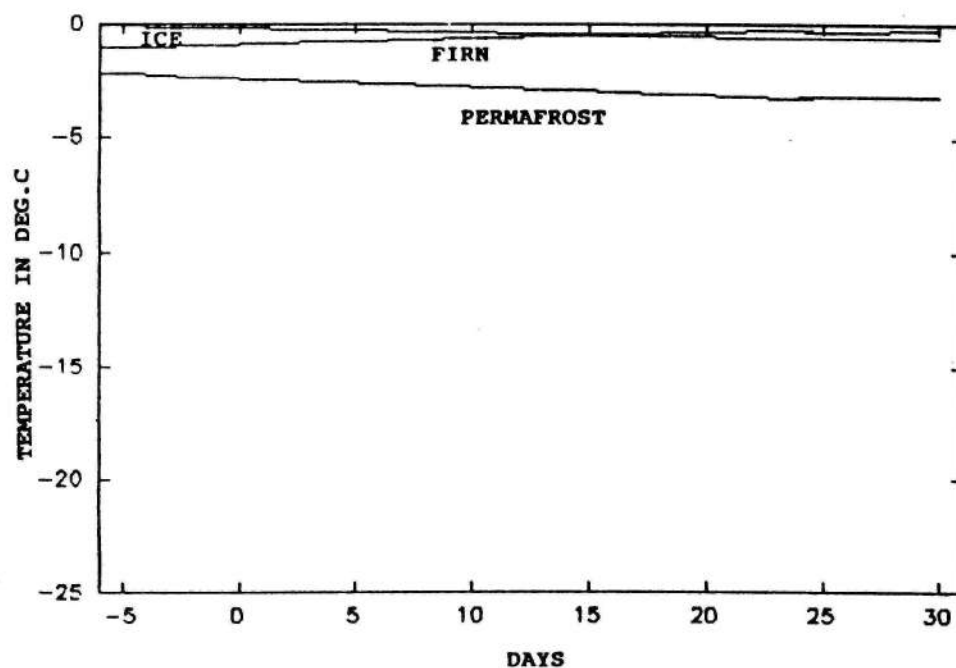


Fig. 9. Thermal plots for ice-firn-permafrost for January 1994

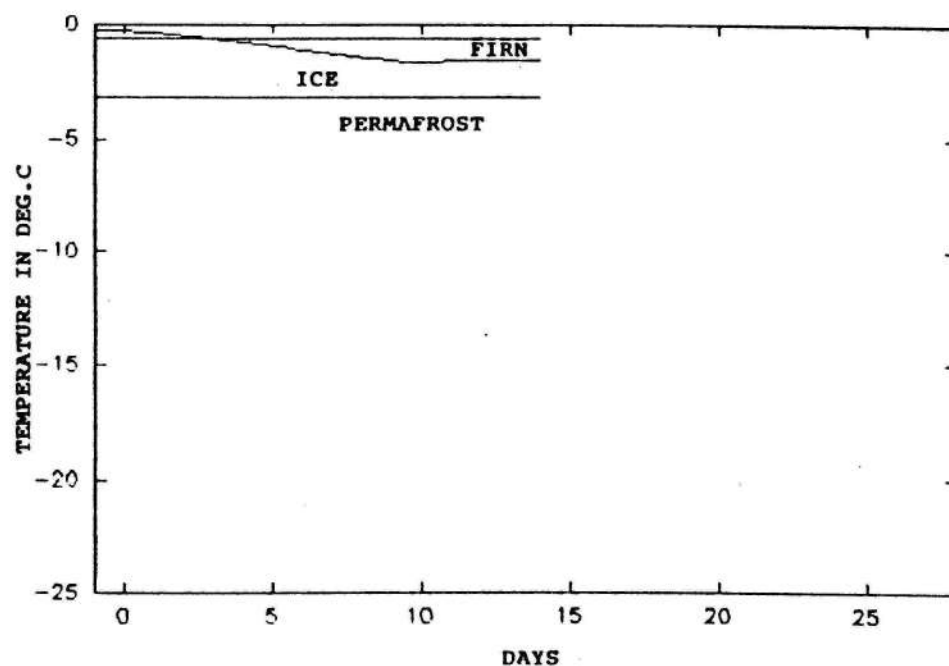


Fig. 10. Thermal plots for ice-firn-permafrost for February 1994

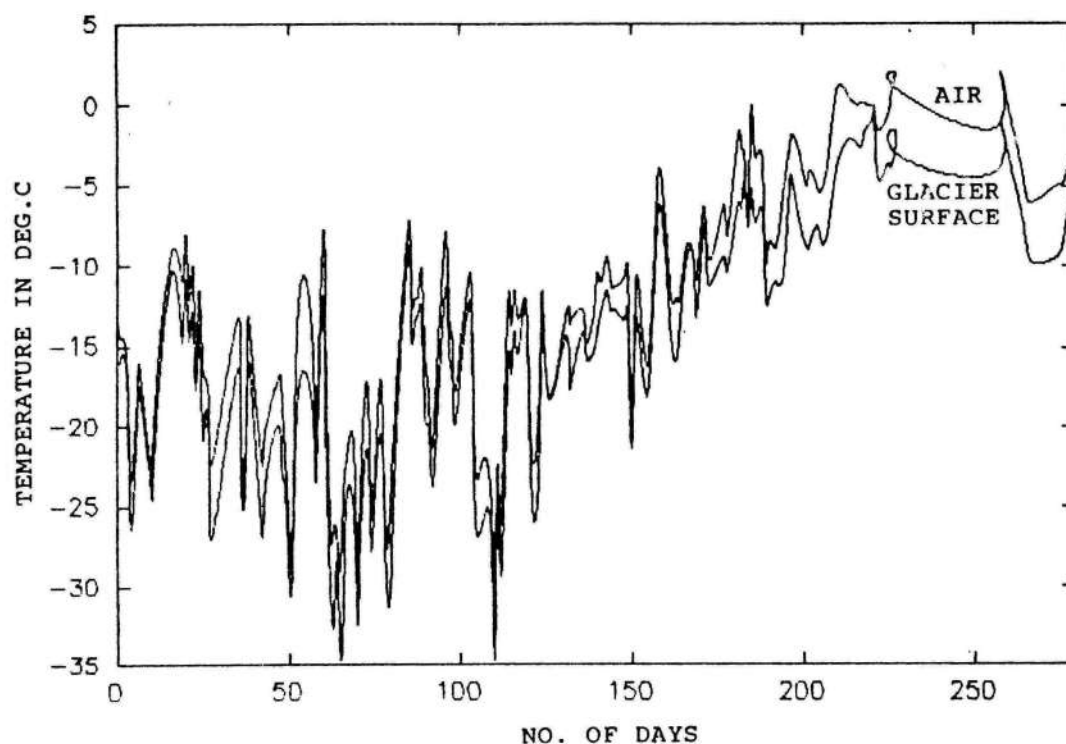


Fig. 11. Comparative overall thermal plots for air temperature and glacier surface temperature

If all the observations of ice-surface temperature and air temperature are compared, a distinct relationship is seen between the two sets. The temperatures on the surface of the glacier are always lower by about 1°C to 5°C (Fig. 11).

The overall comparative thermal plots for these ten months display the thermal amplitudes attained by the three media (Fig. 12).

Conclusions

From the above data and discussion, following inferences may be drawn :

1. Diurnal fluctuations do not affect any of the three selected media at a depth of one metre, or more.
2. Among the three media, blue ice is most conducive to thermal changes. It is the first medium to respond to both the cold front and the warm front.
3. Firn and permafrost have delayed responses to the changing temperatures. Thus, at times, radically opposite trends are displayed by these media, with respect to the trend of surface temperature.
4. The maximum amplitude of temperatures is displayed by blue ice, which ranged from 0°C to -22°C . Firn ranged from 0°C to -15°C . Permafrost showed the minimum amplitude, ranging from -2°C to -14°C .

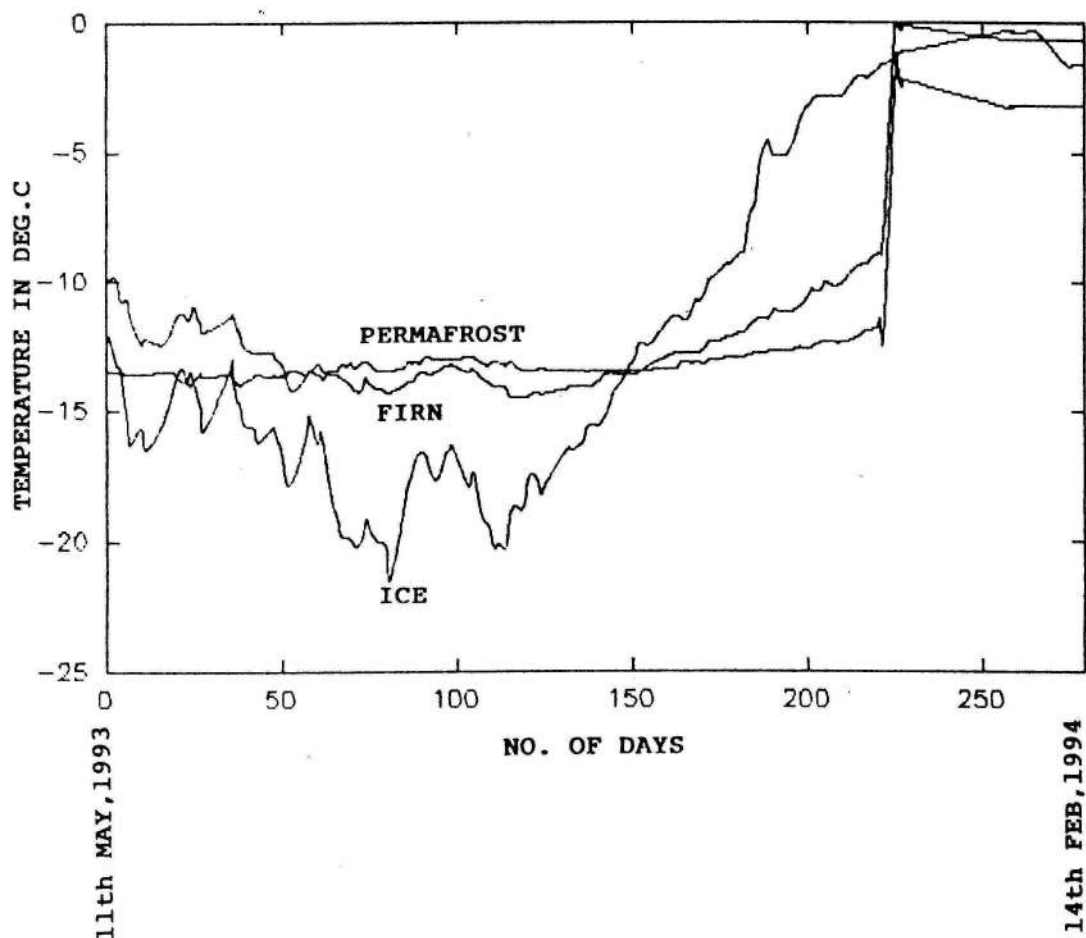


Fig. 12. Comparative overall thermal plots for ice, firn and permafrost from May 1993 to February 1994

5. In December, firn and permafrost both displayed a sudden jump, rising several degrees within a couple of days. This, probably, was due to percolation of warm surface-meltwater to lower strata.
6. The surface of the glacier is always colder than the prevalent air temperature by about 1°C to 5°C .

Part B : Thermal Changes in the Profile of a Crevasse

A narrow hole was dug in the top snow-cover of the crevasse. The thermistor-wire was tied to a small weight and lowered into the crevasse through the hole. The hole was again covered with snow. The thermistor was raised with the help of a pulley and observations were taken at 3 m intervals. Then, it was gradually lowered and repeat observations were taken at 3 m intervals to check the first set of readings.

After the observations, the thermistor was left hanging inside the crevasse; so that the top cover was not disturbed at subsequent visits. Surface-temperature above the crevasse was noted with the help of a normal thermometer.

Observations

Logistic constraints did not permit to visit the crevasse frequently, so only monthly temperature profiles could be drawn. The observations of temperatures recorded at different depths are shown in Table 2 for the months from April to November. In the second half of November, partial surface melting made the crevasse inaccessible and the observations had to be discontinued.

Discussion

Monthwise thermal profiles of the crevasse are displayed in Figs. 13 & 14. These show the relationship between different parts of the crevasse, with variations through different months.

Compared to April, months of May and June show higher overall temperature. This may be explained by higher surface temperatures compared to April. The surface of the glacier became markedly colder in July and August and this is reflected in the overall lower temperatures in the profiles of these months. August has recorded the lowest temperature. In September, the surface temperature has risen and crevasse-profile records a slight overall rise. This trend continues in October and November. It is interesting to note that the profile of November is quite similar to that of April.

Comparing the relationship between different parts of the crevasse, it is seen that below a depth of 15 m the temperature remains almost stable till the

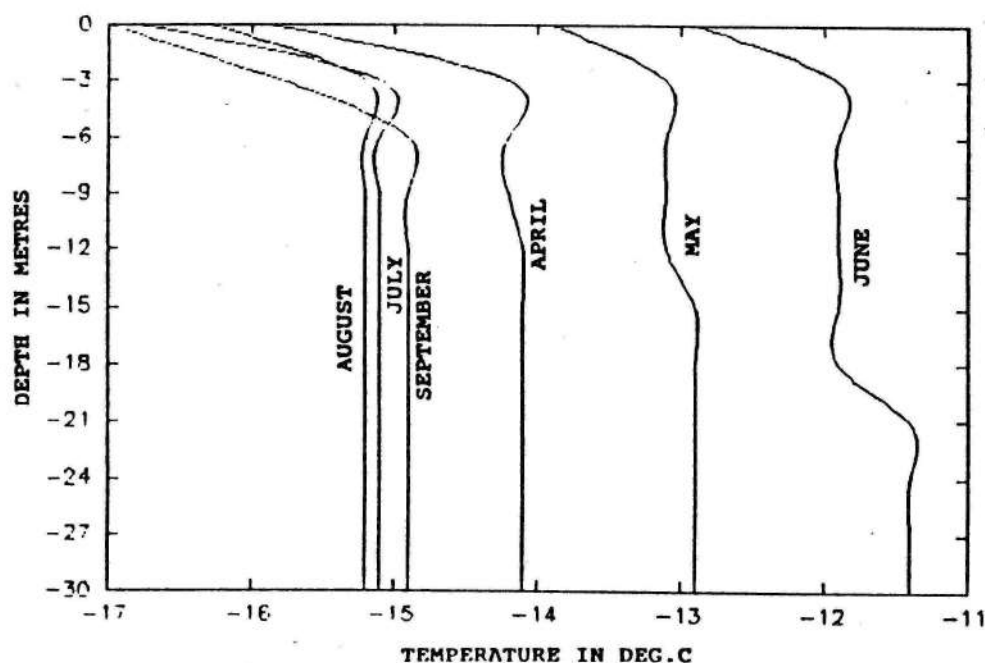


Fig. 13. Monthly thermal profiles of the crevasse for the polar winter months

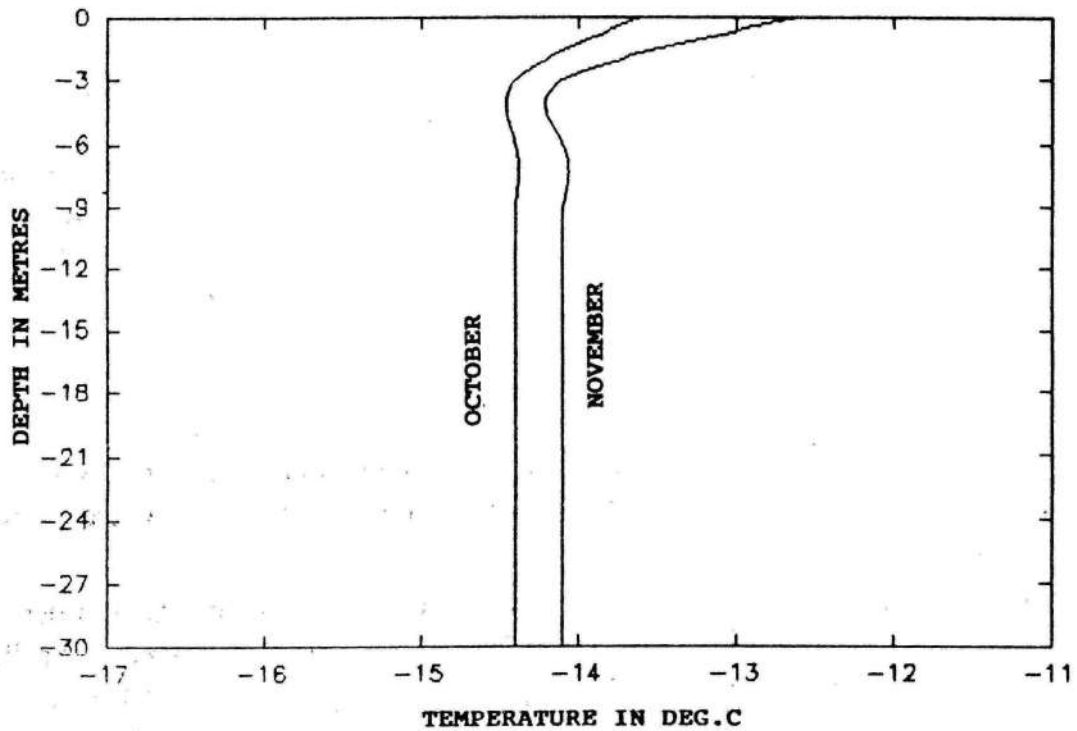


Fig. 14. Monthly thermal profiles of the crevasse for the polar summer months

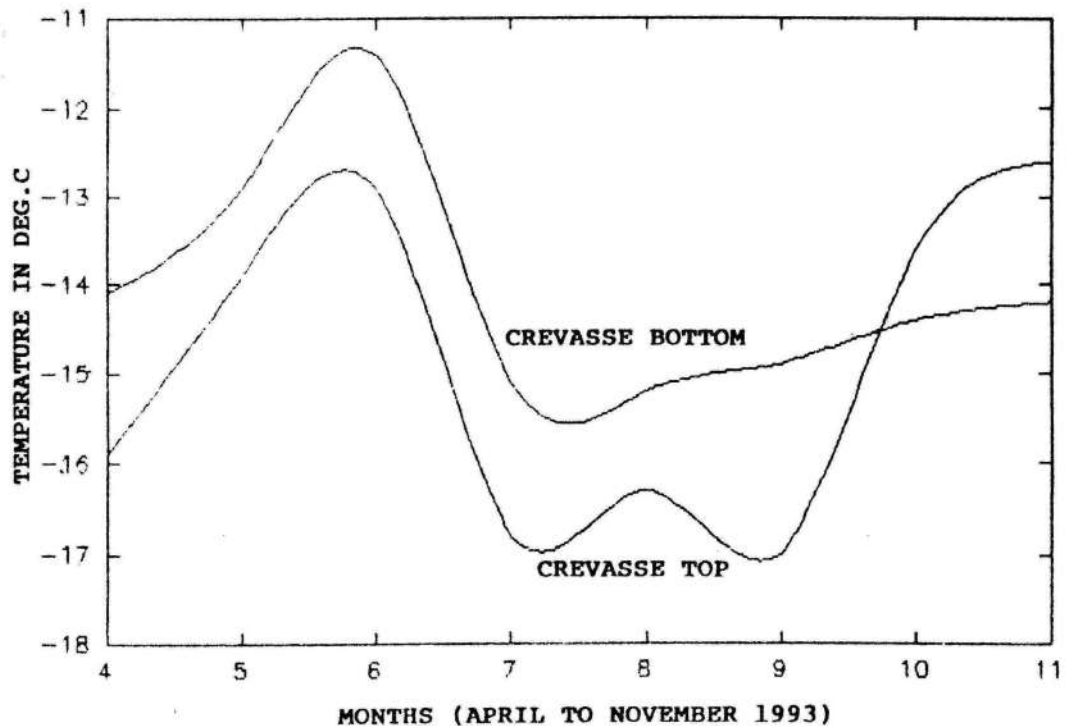


Fig. 15. Monthly thermal profiles of the crevasse from April to November 1993

bottom. Another striking feature is that the bottom of the crevasse is always warmer than the upper parts throughout the polar winter, *i.e.*, from April to September. With the onset of polar summer, there is a temperature inversion inside the crevasse (Fig. 15) and the bottom of the crevasse becomes colder

Table 2: Thermal observations at Veteheia crevasse in 1993

Depth in meters	12th April (°C)	31st May (°C)	29th June (°C)	30th July (°C)	24th August (°C)	28th Sept. (°C)	30th Oct. (°C)	16th Nov. (°C)
surface	-16.8	-9.8	-11.7	-35.5	-32.8	-19.3	-9.5	-5.8
01	-15.9	-13.9	-12.9	-16.8	-16.3	-17	-13.6	-12.6
03	-14.2	-13.1	-11.9	-15.1	-15.2	-15.8	-14.4	-14.1
06	-14.2	-13.1	-11.9	-15.1	-15.2	-14.9	-14.4	-14.1
09	-14.2	-13.1	-11.9	-15.1	-15.2	-14.9	-14.4	-14.1
12	-14.1	-13.1	-11.9	-15.1	-15.2	-14.9	-14.4	-14.1
15	-14.1	-12.9	-11.9	-15.1	-15.2	-14.9	-14.4	-14.1
18	-14.1	-12.9	-11.9	-15.1	-15.2	-14.9	-14.4	-14.1
21	-14.1	-12.9	-11.4	-15.1	-15.2	-14.9	-14.4	-14.1
24	-14.1	-12.9	-11.4	-15.1	-15.2	-14.9	-14.4	-14.1
27	-14.1	-12.9	-11.4	-15.1	-15.2	-14.9	-14.4	-14.1
30	-14.1	-12.9	-11.4	-15.1	-15.2	-14.9	-14.4	-14.1

than the top during October and November. These reversals may be explained by the approach of the cold front and the warm front from the topside of the crevasse and the insulation provided by the depth of the ice.

Yet another significant deduction from the observations is that the temperatures inside a closed crevasse show very minor fluctuations and are quite stable, overall. The bottom of the crevasse varied between -11°C and -16°C only. Though the surface of the glacier became colder than -35°C , the bottom touched a lowest of -15.2°C only. Thus, the inside of a closed crevasse is markedly warmer than the surface of a glacier, during peak polar winter.

Conclusions and Recommendations

The following general inferences may be drawn from the above observations and discussion :

1. The thermal variations inside a closed crevasse are quite small, the bottom of the crevasse varied between -11°C and -16°C only.
2. Below a depth of 15 metres, the temperatures inside a closed crevasse are almost stable till the bottom.
3. The bottom of the crevasse is warmer than the top from April to September. A temperature inversion takes place in October and the bottom becomes colder than the upper parts.

4. The bottom of the crevasse was quite well insulated. Though the surface of the glacier became colder than -35°C , the lowest temperature recorded at the bottom was -15.2°C only.
5. The crevasse shows the lowest overall temperature in the month of August.
6. Monthly profiles are not sufficient to draw dependable inferences and weekly observations should be taken in future, to properly understand the thermal behaviour of a crevasse.

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